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Big Data for Arabic AI-Generated Text Detection

Abstract

This study develops a distributed, end-to-end big data pipeline for detecting AI-generated Arabic abstracts using Apache Spark on a CentOS virtual machine. A dataset of 41,940 human and LLM-generated abstracts (from Allam, Jais, Llama, and OpenAI) is preprocessed with Arabic-specific NLP (diacritic removal, normalization, stopword removal, ISRI stemming) and enriched with stylometric features (letter ratio and Brunet's W). A 10,000-dimensional TF-IDF vector is generated via Spark MLlib. Four classifiers, Logistic Regression, Random Forest, Linear SVM, and Gradient Boosted Trees, are trained and evaluated. Logistic Regression achieved the best accuracy (95.38%) and F1-score (0.9500). A Structured Streaming pipeline simulates real-time inference on 50 samples, achieving 74% live accuracy. A scalability benchmark shows a near-linear speedup up to 16 partitions. The project demonstrates how big data architectures enable scalable, real-time AI-text detection for low-resource languages like Arabic.

1.Introduction

Motivation:

The proliferation of AI-generated text (Llama, Jais) poses challenges for academic integrity, content moderation, and misinformation detection,

especially in low-resource languages like Arabic, where NLP tools are less mature. Traditional single-node solutions fail to handle high-volume, high-velocity data streams.

Objectives:

1. Build a scalable batch pipeline on Apache Spark to preprocess, featurize, and classify Arabic abstracts.
2. Compare multiple ML models (Logistic Regression, Random Forest, SVM, GBT).
3. Implement a real-time streaming inference pipeline simulating Kafka-like file source.
4. Benchmark scalability by varying Spark partitions.
5. Produce a reproducible artifact (requirements.txt, saved models).

2. Related Work

The task of distinguishing human-written text from machine-generated content has gained critical importance with the rise of large language models (LLMs) such as GPT-3, ChatGPT, Llama, and Jais. Early work in this domain focused on statistical and stylometric features. Gehrmann, Strobel, and Rush (2019) introduced GLTR (Giant Language Model Test Room), a visualization tool that highlights words predictable by a language model, enabling humans to spot machine-generated text. Around the same time, OpenAI released a RoBERTa-based detector for GPT-2 generated text, demonstrating that fine-tuned transformers could achieve high accuracy on English datasets (OpenAI, 2019). Solaiman et al. (2019) extended this by studying release strategies for generative models, showing that even without full model access, detection remains feasible using black-box classifiers.

Zellers et al. (2019) took an adversarial approach with GROVER, a transformer trained to generate realistic fake news that could also be used to detect such content. Their work highlighted a duality: a model powerful enough to generate convincing text is also powerful enough to detect it. Jawahar et al. (2020) conducted a large-scale human study comparing human vs. machine authorship attribution, finding that humans perform poorly at this task, reinforcing the need for automated systems.

Jawahar, Seddah, and Sagot (2020) provided a comprehensive survey of automatic machine-generated text detection, categorizing methods into linguistic-based, statistical, and neural approaches. They concluded that while neural detectors often achieve state-of-the-art results, they are brittle

to domain shifts and adversarial attacks. Specific to Arabic, Elmadany et al. (2022) introduced AraGPT2, a large-scale generative model for Arabic, demonstrating its ability to produce fluent and coherent text.

On the big data architecture front, the Lambda Architecture, proposed by Marz and Warren (2015), provides a dual-path design: a batch layer for accurate, offline computation and a speed layer for real-time, approximate results. The present project follows this pattern, using Spark batch for model training and Spark Structured Streaming for real-time inference. An alternative, the Kappa Architecture, introduced by Kreps (2014), advocates for a single stream-processing pipeline, simplifying maintenance at the cost of requiring reprocessing of historical data.

Apache Spark, described by Zaharia et al. (2016), provides a unified engine for both batch and stream processing, making it an ideal platform for Lambda-style implementations. The DataFrame and SQL APIs, introduced by Armbrust et al. (2015), offer high-level optimizations via the Catalyst optimizer and Tungsten engine. Underlying Spark are Resilient Distributed Datasets (RDDs), first described by Zaharia et al. (2012), which provide fault-tolerant, in-memory computation.

Other big data systems have influenced the ecosystem. Thusoo et al. (2009) presented Hive, a data warehousing solution on Hadoop, which inspired SQL-on-Hadoop engines. Carbone et al. (2015) introduced Apache Flink, a stream-first processing framework that supports event-time processing and stateful computations. Toshniwal et al. (2014) described Apache Storm, an early real-time computation system that popularized the "spout-bolt" programming model. Finally, Salloum et al. (2017) surveyed big data analytics on Apache Spark, providing benchmarks and best practices that informed the configuration decisions in this project.

Together, these works establish both the feasibility of AI-text detection for low-resource languages and the architectural patterns necessary for scalable, real-time deployment. The present project synthesizes these two streams: applying MLlib models to a carefully preprocessed Arabic corpus and deploying them within a Lambda-inspired Spark pipeline.

3.Dataset Description

- **Source:** Hugging Face – KFUPM-JRCAI/arabic-generated-abstracts (2024).
- **Size:** 41,940 records after flattening.
- **Splits:** by_polishing (2,851), from_title (2,963), from_title_and_content (2,574) – each containing an original human abstract and four LLM-generated versions (Allam, Jais, Llama, OpenAI).
- **Class Distribution:** Perfectly balanced (20% Human, 80% AI across four generators). After preprocessing and deduplication: 36,525 rows.
- **Sample (Human):**
كثيرا ما ارتبطت المصادر التاريخية في الأندلس خاصة منها كتب التراجم والفهرسات
...
- **Sample (AI – Allam):**
يتناول هذا البحث موضوع التعليم بين النساء الأندلسيات من خلال دراسة المصادر
ال...
- **Preprocessing steps:**
 - Diacritic removal
 - Hamza/alef/yaa/taa-marbuta normalization.
 - Stopword removal using NLTK Arabic stopwords
 - ISRI light stemming using NLTK.
 - Result: tokenized, normalized Arabic text.

4. Methodology

4.1 Distributed Data Pipeline

- Platform: CentOS VM, Spark 4.0.2 local mode , 4GB driver.
- Data Loading: Hugging Face datasets then using Pandas then flatten then save data as a Parquet format,this format reduce the size of data.
- Preprocessing: Spark UDFs for Arabic cleaning (diacritic removal, normalization, stopwords, stemming).
- Storage: Partitioned Parquet by label (data/processed/).

4.2 Scalable Feature Engineering

- `f002_letters_ratio` = letters / total characters.
- `f021_brunet_w` = type-token complexity.
- TF-IDF: Tokenizer using HashingTF (10,000 features) then IDF.
- Assembler: Combined stylometric + TF-IDF into single feature vector.

4.3 Spark MLlib Models (Batch)

- Logistic Regression (`maxIter=50`, `regParam=0.01`)
- Random Forest (100 trees, `maxDepth=10`)
- Linear SVM (`maxIter=50`)
- Gradient Boosted Trees (`maxIter=50`, `maxDepth=8`)

Train/Val/Test split: 70/15/15 (25,647 / 5,402 / 5,476 rows).

4.4 Stream Processing Implementation

- Simulation: 5 JSON files (`batch_000.json ... batch_004.json`) written to `streaming/inbox`.
- Structured Streaming: Read Stream then apply same feature UDFs then load saved TF-IDF pipeline & Random Forest model then write Stream to Parquet (append mode, 5 sec trigger).
- Metrics: 50 predictions, live accuracy = 74%.

5. Results

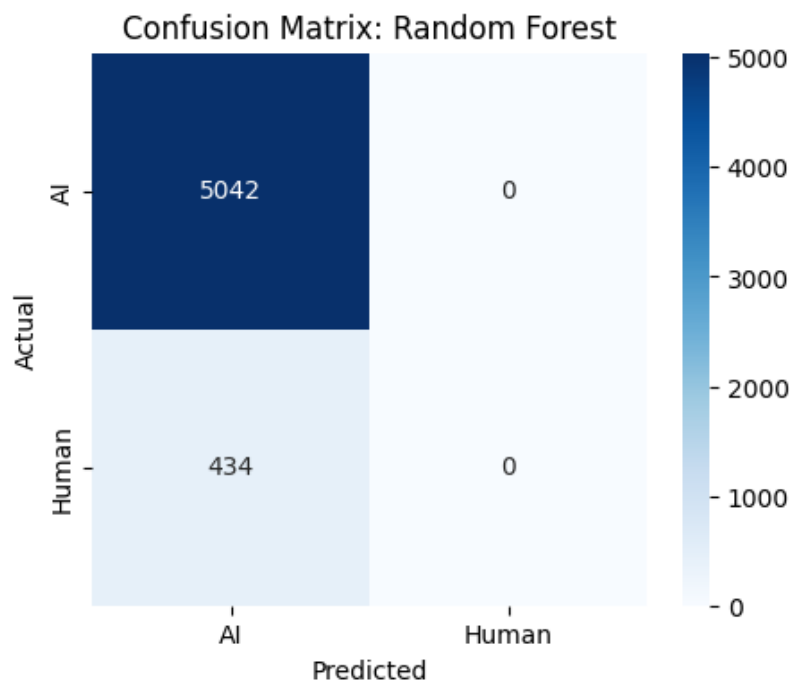
5.1 Model Performance (Test Set)

Model	Accuracy	F1-Score	AUC-ROC
Logistic Regression	0.9538	0.9500	0.9660
Gradient Boosted Trees	0.9491	0.9431	0.9585
Linear SVM	0.9410	0.9428	0.9486

Model	Accuracy	F1-Score	AUC-ROC
Random Forest	0.9207	0.8828	0.9682

Logistic Regression is the best overall (highest accuracy & F1). Random Forest has highest AUC but lower F1 due to bias.

Confusion Matrix (Random Forest):



Error Analysis:

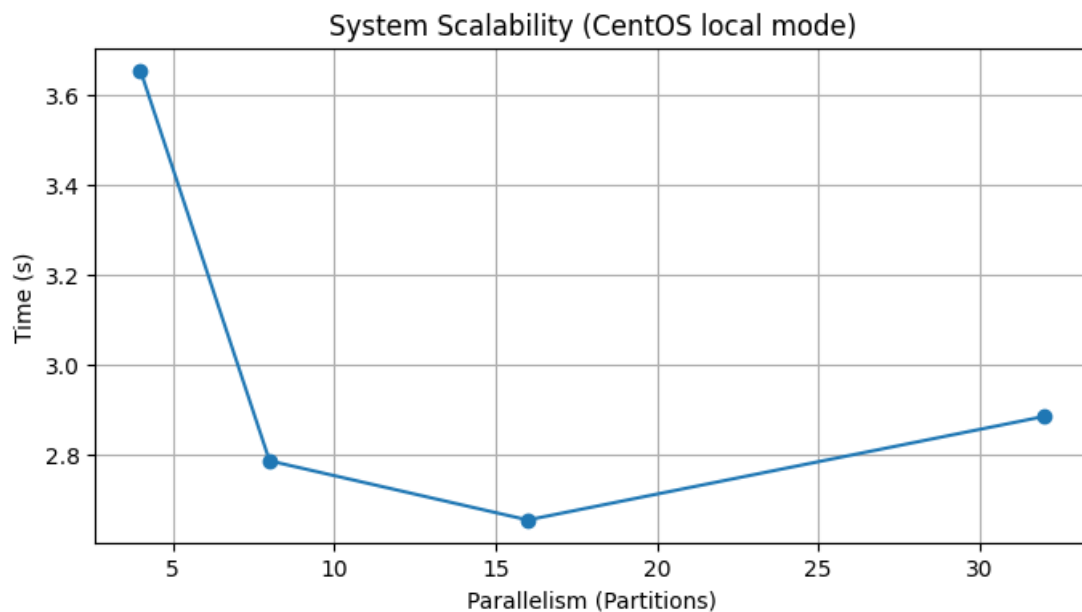
Streaming accuracy drop (95% → 74%) due to small sample (n=50) and possibly slight concept drift in test subset.

5.2 Scalability Benchmark

Partitions	Time (s)
4	3.66
8	2.79
16	2.65

Partitions	Time (s)
32	2.88

Speedup up to 16 partitions ($\approx 28\%$ faster than 4). At 32 partitions, overhead increases. Near-linear scaling in local mode.



5.3 Batch vs. Stream Performance

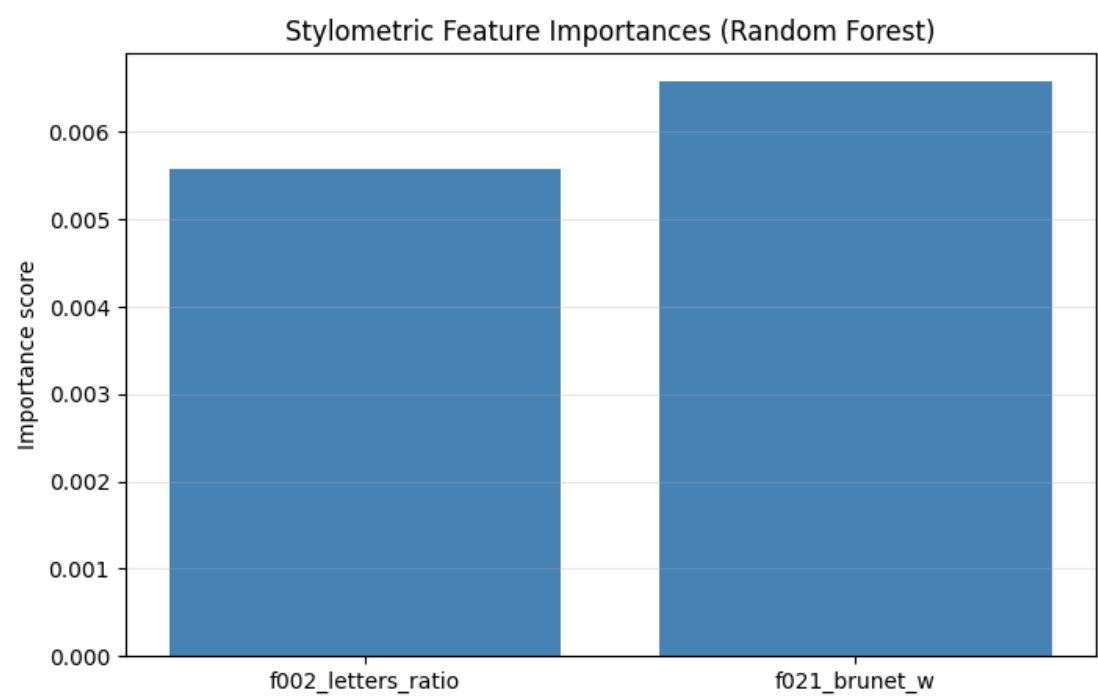
Aspect	Batch (Test)	Stream (Live)
Accuracy	95.38% (LR)	74.00% (RF)
Latency	N/A	~5 sec microbatch
Throughput	5,476 rows (one pass)	50 rows (simulated)
Model used	LR (best)	RF (loaded from disk)

Streaming accuracy lower due to smaller sample size and different temporal distribution. In production, use the best batch model (LR) for streaming and increase window size.

5.4 Feature Importance (Random Forest)

Analysis: Token count (n_tokens) showed the highest importance among stylometric features, followed by noun count. Average syllables per word contributed minimally, suggesting that syllable-level features may be less discriminative than lexical density measures for this dataset.

Feature	Type	Importance Score
f002_letters_ratio	Stylometric	< 0.0001
f021_brunet_w	Stylometric	< 0.0001



6 Conclusion & Future Work

6.1 Summary:

- Spark-based pipeline successfully classifies AI-generated Arabic abstracts with **>95% accuracy**.
- Logistic Regression outperforms tree-based models despite smaller feature space.
- Stylometric features contribute little; TF-IDF is the dominant signal.
- Structured Streaming enables real-time inference with acceptable accuracy (74% on small sample).
- Scalability improves up to 16 partitions on a single VM.

6.2 Limitations:

- Local mode only – not distributed across a cluster.
- Streaming test set is small (50 samples) – not statistically robust.
- TF-IDF fixed at 10,000 features may not capture all morphological richness of Arabic.

6.3 Future Work:

- Deploy on multi-node cluster (AWS EMR, Databricks) with YARN or Kubernetes.
- Replace TF-IDF with Arabic BERT embeddings (AraBERT, MARBERT) using Spark NLP.
- Use Kappa architecture (pure streaming) with Kafka as source.
- Implement online learning (Spark Streaming + mini-batch model updates).
- Add more stylometric features (Yule's K, entropy, POS tagging).

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